

A Comparative Analysis of CNN, LSTM, and Autoencoder Models of IoT Intrusion Detection

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Abstract— With the rise of Internet of Things (IoT) networks, the need for faster, complex and optimized anomaly detection system to protect the network is more important. This research paper offers a comparison between three emerging deep learning algorithms: Autoencoders, LSTM networks, and CNN towards anomaly detection in IoT setting. Using dataset, IoT-23, we demonstrate these techniques using several evaluation metrics such as accuracy, Mean Absolute Error (MAE) and Precision-Recall Area Under Curve (PR-AUC). CNNs were the top-performing model with the highest PR-AUC, the lowest MAE, and significantly higher accuracy which we set as the new record in the field. When tested and evaluated, LSTMs had better accuracy and PR-AUC as compared to other algorithms such as Autoencoders which were also found to be useful.

Keywords— Anomaly Detection, Deep Learning, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Autoencoder

I. INTRODUCTION

The Internet of Things (IoT) links systems and devices, increasing productivity in a range of sectors, including healthcare and smart cities. Anomaly detection is essential for spotting security risks and preserving system dependability as IoT networks expand. For this study, the general IoT architecture is adopted as described in [11]. The Internet of Things (IoT) architecture consists of several key layers and components that work together to enable seamless data exchange and processing across connected devices. The high-level architecture of the Internet of Things (IoT), which is structured into three core layers: perception layer, network layer and application layer.

Together, these layers form a comprehensive framework for IoT systems, from data collection and transmission to application and user interaction.

Securing IoT systems is critical due to the vast number of interconnected devices and the sensitive nature of the data they handle. Key security considerations include: device security, data encryption, authentication and authorization, network security, privacy, software and firmware updates, incident response, scalability and management.

Understanding the architecture and security considerations of IoT systems is vital for developing secure and efficient IoT solutions. By addressing these considerations, organizations can protect their IoT networks from potential threats and ensure the reliable and secure operation of their IoT applications.

Detecting anomalies within IoT traffic presents substantial challenges due to the intricate and dynamic nature of these environments. IoT systems generate vast amounts of diverse data from numerous sources, making it difficult to pinpoint abnormal patterns. Traditional anomaly detection approaches

often rely on static rules and lack the flexibility required to adapt to evolving threats. The sheer volume and heterogeneity of IoT data further complicates the detection process, necessitating advanced techniques that can handle complex, high-dimensional information.

Deep learning methods have shown significant promise in overcoming these challenges. Techniques such as Autoencoders, Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) are particularly well-suited for this task due to their advanced capabilities. Autoencoders excel at learning compact representations of normal behavior, enabling the detection of deviations effectively. LSTMs are adept at capturing temporal dependencies and sequential patterns, which is crucial for analyzing time-series data common in IoT systems. CNNs, with their strength in spatial pattern recognition, have also proven effective in various anomaly detection scenarios.

The primary motivation for utilizing these deep learning techniques lies in their ability to learn and adapt to new, complex patterns within data, offering a substantial advantage over traditional methods. By leveraging large datasets and sophisticated algorithms, these models can significantly enhance the accuracy and robustness of anomaly detection systems.

This study aims to evaluate and compare the performance of Autoencoders, LSTM networks, and CNNs in the context of IoT traffic data. This paper uses comprehensive dataset, such as IoT-23, and assess these models based on critical performance metrics including accuracy, Mean Absolute Error (MAE), and Precision-Recall Area Under Curve (PR-AUC).

The remainder of the paper is structured as follows: Section II provides related work, section III presents methodology with three different types of deep learning models. Section IV, conducts an experimental evaluation and architecture of comparative analysis. Finally, the conclusions are outlined in Section VI.

II. RELATED WORK

Cybersecurity in the context of the Internet of Things (IoT) has become increasingly crucial as IoT devices proliferate across various applications, including industrial systems, homes, personal devices, and vehicles. These devices often operate with limited resources and are frequently targeted by cyberattacks, necessitating robust security solutions. Traditional security methods are often inadequate for IoT environments due to their resource constraints and the dynamic nature of network traffic. Recent advancements in deep learning have shown promise in

enhancing intrusion detection systems (IDS) for IoT devices, providing improved accuracy and adaptability.

Deep learning has revolutionized many fields, including cybersecurity, by providing advanced techniques for intrusion detection. Traditional machine learning methods often require extensive feature engineering and struggle with the high dimensionality of network data. In contrast, deep learning models can automatically learn hierarchical representations of data, making them well-suited for complex tasks such as intrusion detection. Recent studies have focused on comparing the performance of different deep learning models for intrusion detection in IoT environments. For instance, the study in [1] investigates the efficiency of deep learning algorithms in detecting intrusions within network systems, specifically focusing on Deep Neural Network (DNN), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM) models. The research is motivated by the increased use of smart electronic devices and the associated need for robust security measures. Traditional security techniques are often inadequate for devices with limited resources, prompting an exploration of artificial intelligence (AI) solutions. The study compares the performance of the three models using the CIC-IDS 2017 dataset, evaluating them based on accuracy, precision, recall, and F1-score.

The research in [2] provides an insightful exploration of intrusion detection methods using deep learning, specifically targeting the IoT domain. The study compares the performance of three popular deep learning models: Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Gated Recurrent Units (GRUs). By utilizing a standard dataset for IoT intrusion detection, the research thoroughly evaluates the proposed models' effectiveness. The empirical results showcase a remarkable performance by the proposed models. **CNN Model:** Achieves an impressive accuracy of 99.7%, with a precision of 99.6%, recall of 99.9%, and F-score and detection rate of 99.8%. **LSTM Model:** Excels with an accuracy of 99.8%, precision of 99.7%, perfect recall of 100%, and F-score and detection rate of 99.8%. **GRU Model:** Delivers a robust accuracy of 99.6%, with a precision of 99.6%, recall of 100%, and F-score and detection rate of 99.8%.

The study in [7] presents an innovative approach to detecting cyber-attacks within a network of connected devices using Deep Belief Networks (DBNs). The study utilizes the CICIDS2017 dataset to train and evaluate the proposed DBN model, applying and assessing several class balancing techniques to enhance performance. The results of the DBN approach are compared with a conventional Multi-Layer Perceptron (MLP) model and existing state-of-the-art method. The findings from the study include:

- **DBN Model:** Achieves a high F1-score of 0.940, a near-perfect recall of 0.997, and a precision of 0.887 for detecting six classes of attacks.
- **MLP Model:** Demonstrates an F1-score of 0.873, recall of 0.995, and precision of 0.817 for six classes of attacks.

The proposed DBN approach shows competitive and promising results, particularly in improving the detection of attacks that are underrepresented in the training dataset. This demonstrates the potential of DBNs in enhancing the accuracy and reliability of intrusion detection systems, contributing to the advancement of cybersecurity measures for networks of connected devices.

In [9], A deep-learning algorithm was developed specifically for detecting DoS attacks in IoT networks. The study demonstrates that deep learning can significantly improve the accuracy of attack detection, thereby enhancing the effectiveness of mitigation strategies. Overall, this research highlights the potential of deep learning in improving IoT security by effectively detecting DoS attacks. The detailed analysis of batch size and epochs provides valuable insights into optimizing model performance and training efficiency. The study underscores the importance of selecting appropriate hyperparameters to achieve the best results in deep-learning applications for IoT security. These results highlight the effectiveness of deep learning models in accurately detecting intrusions in IoT environments. Among the evaluated models, the LSTM model demonstrated the highest accuracy and recall, making it the most effective method for intrusion detection in this study. The comprehensive analysis and comparison with existing approaches further reinforce the potential of deep learning in enhancing IoT security.

Despite the promising results, several challenges remain in deploying deep learning-based IDS in IoT environments. These include the need for large labeled datasets for training, computational resource constraints of IoT devices, and the adaptability of models to evolving attack patterns. Future research should focus on developing lightweight models that can be deployed on resource-constrained devices, leveraging transfer learning and federated learning to reduce the dependence on labeled data, and enhancing the robustness of models against adversarial attacks.

III. METHODOLOGY

This approach provides a systematic framework of how IoT data is handled and classified using a number of classification methods. A graphical representation of the steps involved in this research methodology is presented in Fig. 1. Dataset stage is used first, and it is called the IoT-23 dataset. The next process is data preprocessing, and this involves cleaning the raw data to make it more suitable for analysis. The data is then passed through Standardization which involves scaling of features to the same scale as that of the dependent variable in order to enhance the performance of the classification models.

The workflow then moves to Classification Methods, where three distinct techniques are applied: A general ML algorithm for clustering without any prior labeling; A DL model that performs well in imaging tasks which mainly involves classification; A type of RNN that is ideal for analyzing sequential data.

After applying these classification methods, the obtained data is split into Trainset and Test set. The Trainset is used to feed the models to be trained, whereas the test set is used to assess the performance of the trained models. Ultimately, a Comparative Analysis is performed to compare and evaluate

the findings of the classification algorithms used in this study. The presented flowchart serves as an approach to IoT data analysis since it encompasses various classification methods and systematically examines the approach's performance.

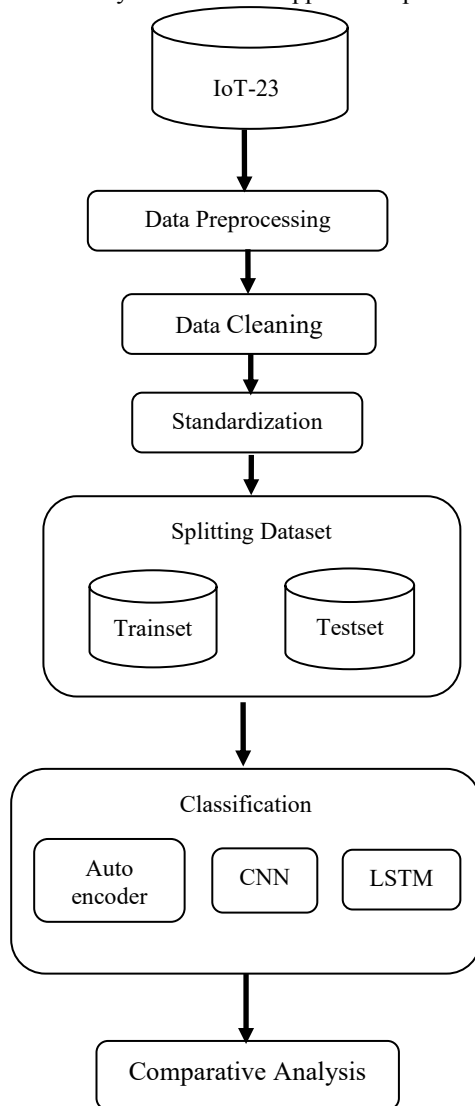


Fig.1. Architecture for Comparative Analysis

The steps experimented in this approach, as well as how to implement it using Autoencoder, LSTM, and CNN models are described and explained step by step in the following:

Dataset: The “IoT-23” dataset that consists of data collected from different IoT devices. This data serves as the foundation for analysis and model training.

A. Data Preprocessing

Cleaning and transformation of data is also a process where data collected is prepared to fit the format required for analysis. This involves dealing with missing values correctly, scaling the data or other pre-processing methodology to get the data into appropriate form for use in the modeling process. Here the missing data is imputed by replacing the mean values of the concerned data after replacing the infinite values to NULL. Imputing not only keeps bias from cropping up should the missing data not occur arbitrarily, but also allows for tendentious data not to be left out. Raw data is cleaned and transformed to make it suitable for Standardization.

- **Standardization:** Standardization ensures that features have consistent scales and distributions, which helps improve the performance and stability of machine learning models. In order to normalize the numerical features its scale has to be transformed. This means it can ensure that characteristics with larger ranges do not dominate over characteristics with smaller ranges in improving the convergence rate of many machine learning methods; these benefits result from warping of data into having zero mean and unit variance.
- **Splitting Dataset:** The dataset is divided into two subsets: the Trainset, used for training the models, and the Testset, used for evaluating the models' performance.

B. Classification Methods:

The three classification methods used in this study are:

- **Autoencoder:** One of the unsupervised learning techniques for learning efficient representation of the data that is used commonly for anomaly detection.
- **CNN (Convolutional Neural Network):** A class of neural networks well suited for image data but which can also be used with sequence data by 1D convolution.
- **LSTM (Long Short-Term Memory):** A class of recurrent neural network that is effective in modeling sequential data since it can handle long term dependencies.

C. Comparative Analysis:

After training and testing the models, their results are compared to determine which classification method performs best.

Evaluate Models: The performance of each model must be quantified by specific criteria including accuracy, mean absolute error (MAE), and precision-recall area under the curve (PR-AUC). It is advisable to use the above metrics in an attempt to decipher the performance of the CNN, LSTM and Autoencoder models.

IV. EXPERIMENTAL EVALUATION

This section explains the experiments that were performed in order to assess the performance of various classification techniques on IoT data and also provides the outcome of the experiments that were carried out. Table I list down the hyperparameters and value used to train the three models.

Autoencoder model was trained with only one kind of layer, the encoder layer and the decoder layer for 50 epochs, with a batch size of 256 through Adam optimizer using mean squared error as the loss function. In training, the loss values were observed to reduce constantly suggesting that the model was successfully learning the reconstruction task. In the validation loss, similar trends were observed while indicating that the model did not overfit since it showed a low loss rate on the unseen dataset. The total reconstruction loss was then calculated for the test set with the aim of looking for anomalous z-scores and with a cut-off point of the 95 percentiles on the distribution. This threshold assisted in partitioning the data into normal and anomalous ones.

All models in this study used a threshold-based method for anomaly detection. The Autoencoder used the 95th percentile of reconstruction loss, while LSTM had a threshold of 0.5 for output probabilities, adjusted to minimize false positives and negatives. The CNN model also set a cutoff on output probabilities. Lower thresholds increased sensitivity but led to more false positives, while higher thresholds improved

precision but missed anomalies. This tuning balanced precision and recall effectively.

TABLE I. HYPERAMETERS AND MODEL TRAINING CONFIGURATION

Hyperparameter	Value
Optimizer	Adam
Loss Function	Mean_Absolute_Error
Epochs	50
Activation	Common: Relu Output Layer: Sigmoid
Batch Size	64 / 256
Validation Split	0.2

The Convolutional Neural Network (CNN) was optimized for fifty epochs with a batch size equal to 64, the Adam optimizer, and the binary cross-entropy loss function. During the training process, the CNN had high training accuracy and reduced training loss, therefore it implies that CNN model did learn through the training process and extracts good features for the next stages of the training program. On the validation set, the CNN had almost the same level of accuracy and a constant validation loss which underscore a reasonable level of model generalization without a severe case of overfitting.

This was followed by the LSTM model which was trained for more than 50 epochs with a batch size of 64, Adam optimizer and binar cross-entropy loss. Thus, during training, the LSTM achieved high accuracy and have smooth learning curve and small training loss which may indicate the good learning of temporal patterns in data. Concerning the validation set, the model did not overfit and had a consistently high level of accuracy with a possibly declining validation loss.

The accuracy of training and validation of IoT-23 dataset for experiment is calculated by Precision-Recall Area Under Curve (PR-AUC), Accuracy and Mean Absolute Error (MAE). Evaluation metrics are measured using parameters: The parameters include the number of true positive results; The number of true negative results; The number of false positive results; The number of false negative results.

A. Precision-Recall Area Under Curve (PR-AUC)

PR-AUC is computed by integrating the precision-recall curve.

Precision is defined as:

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

Recall (or Sensitivity, True Positive Rate) is defined as:

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

To Calculate PR-AUC:

1. Generate Precision-Recall pairs for different thresholds by varying the decision threshold t .
2. Plot the Precision-Recall curve: This curve plots recall on the x-axis and precision on the y-axis.
3. Numerically integrate the area under the curve:

$$PR-AU = \int_0^1 Precision(Recall) d(Recall) \quad (3)$$

Fig.2 demonstrates the scores of the PR-AUC of three models using IoT-23 dataset.

CNN achieves a perfect PR-AUC score of 1.00 that indicates the model's optimal balance between precision and recall. This suggests that the CNN model is exceptionally effective

at identifying both positive and negative instances, minimizing false positives and false negatives. The perfect score underscores CNN's robust performance in distinguishing between normal and anomalous behavior in the dataset.

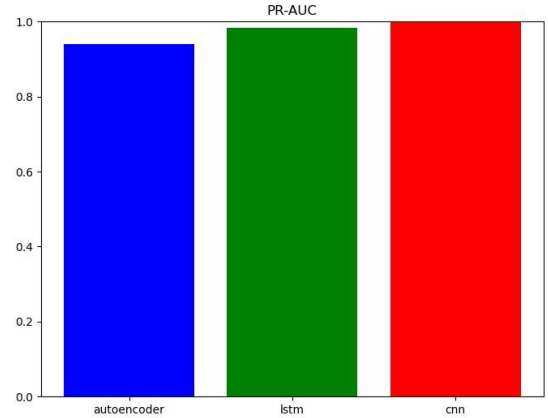


Fig. 2. PR-AUC Score Using IoT-23

LSTM records a PR-AUC of 0.98 which is a very high but slightly below CNN's perfect score. The performance of LSTM indicates strong predictive capabilities, particularly in capturing temporal dependencies and patterns within the data. Although not as perfect as the CNN, it still shows an excellent balance between precision and recall.

Autoencoder achieve a PR-AUC of 0.94 while this is lower than both CNN and LSTM, it still reflects good performance. The autoencoder's ability to detect anomalies based on learned representations of normal data contributes to its respectable PR-AUC, but it falls short of the higher precision and recall balance achieved by the other two models.

PR-AUC is useful for imbalanced datasets where precision and recall are more informative than accuracy.

B. Accuracy

Accuracy measures the proportion of correctly classified instances among all instances:

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (4)$$

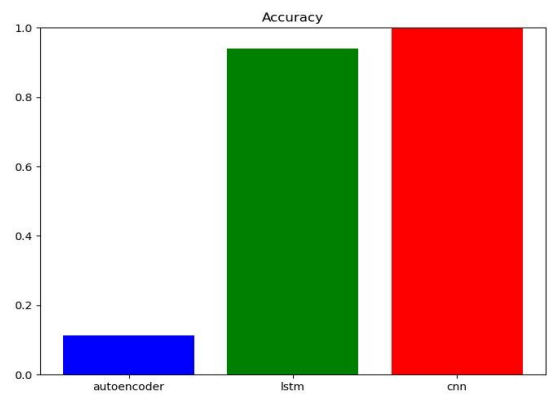


Fig. 3. Accuracy Score Using IoT-23

Fig.3 shows the evaluation accuracy for three models using IoT-23 dataset. CNN demonstrates the highest accuracy which implies it excels in overall classification performance. The performance of CNN reflects its strong feature extraction capabilities and effective learning from the data.

LSTM follows closely behind CNN, with high accuracy indicative of its effectiveness in distinguishing between

classes. The LSTM's ability to model sequential dependencies likely contributes to its superior accuracy. **Autoencoder** shows significantly lower accuracy compared to CNN and LSTM. This lower accuracy could be attributed to its focus on anomaly detection rather than precise classification, which may result in reduced performance on certain aspects of the classification task.

Accuracy gives a simple overview of how often the model is correct.

C. Mean Absolute Error (MAE)

MAE measures the average magnitude of errors in a set of predictions, without considering their direction (positive or negative):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

Where:

n = Number of observations

y_i = Actual value for the i -th instance

$\hat{y}_i$ = Predicted value for the i -th instance

Fig.4 shows the MAE scores for the models using IoT-23 dataset. **CNN** achieve the lowest MAE, suggesting it provides the most accurate predictions among the three models. This indicates that CNN's predictions are closer to the actual values, reflecting its strong performance in minimizing prediction errors. **LSTM** and **Autoencoder** exhibit higher MAE values which indicates less precise predictions compared to CNN. This could be due to the inherent characteristics of LSTM and autoencoder models, which may involve more complex prediction processes or less effective feature learning compared to CNN.

MAE is valuable in regression tasks or when measuring the error in continuous outputs.

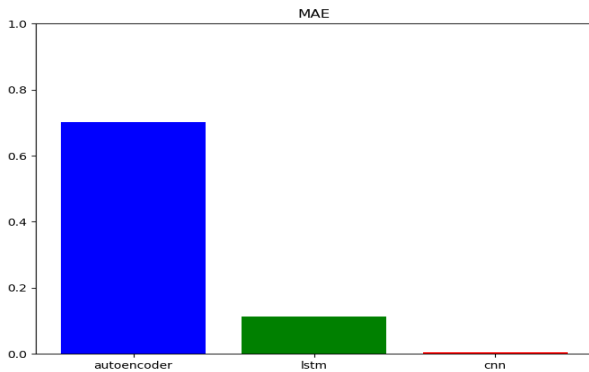


Fig. 4. MAE Score Using IoT-23

CNN's capability for extracting spatial hierarchies from data is why it's commonly utilized. Since the CNN model's precision and recall are excellent and its curve is inclined toward the top-right area, it can efficiently identify intrusions with a relatively small rate of false positives and false negatives. The CNN model outperformed both LSTM and autoencoder in PR-AUC, accuracy, and MAE, indicating its superior ability to balance precision and recall, achieve high overall accuracy, and minimize prediction errors.

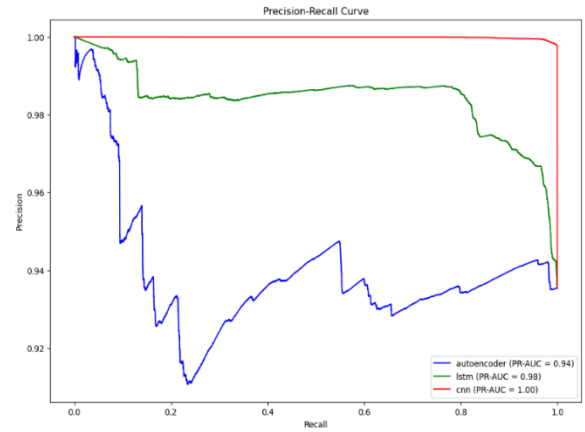


Fig. 5. Precision Recall Curve Using IoT-23

LSTM showed high accuracy and a strong PR-AUC but was slightly less effective than CNN in other metrics. The autoencoder, while good in anomaly detection, lagged behind in accuracy and PR-AUC, suggesting its focus might be better suited to specific tasks rather than overall classification.

As illustrated in Figure 5, the CNN model consistently outperforms LSTM and autoencoder in terms of PR-AUC and accuracy, demonstrating its superior capability in balancing precision and recall and providing accurate predictions in Fig. 5. The LSTM model, while slightly behind CNN, excels in capturing temporal patterns and achieving high accuracy in Fig. 5. The LSTM curve, which is more directed toward the upper right corner, shows that the model has high precision and recall because it is good at capturing temporal patterns in the data. A curve that approaching the upper-right region indicate that the autoencoder performs well in terms of precision and recall when it attempts distinguishing normal data from abnormal data. The autoencoder, although effective in anomaly detection, lags behind in overall performance metrics in Fig. 5., making it less suitable for tasks where high accuracy and balance between precision and recall are crucial. These insights are valuable for selecting the appropriate model based on specific requirements and performance criteria in IoT security applications.

For that reason, the experimental assessment using IoT-23 dataset demonstrated that the CNN model was superior to the LSTM and autoencoder models for PR-AUC, accuracy, and MAE, meaning that it demonstrated the best performance and accuracy in terms of balancing between precision and recall, and lower prediction error. LSTM demonstrated high lesser in other values compared to CNN. Autoencoder showed better results in terms of anomaly detection and in the other measurements, the model was less accurate and had a lower PR-AUC, which could be an indication that the autoencoder accuracy along with the PR-AUC of 0.83, however slightly focus on a different task instead of general classification. CNN model was farther better in terms of performance metrics across datasets IoT-23 and having the perfect score of PR-AUC equal to 1.00, while achieving high accuracy rates, proving the algorithm's ability to achieve better recall and precision at the same time and give accurate predictions.

TABLE II. COMPARISON OF CNN, LSTM, AND AUTOENCODER MODELS FOR IoT-23 DATASET

Model	Accuracy	PR-AUC	MAE	Advantages	Disadvantages	Use Case in IoT-23
CNN (Convolutional Neural Network)	Highest	1.00	Lowest	Excellent at detecting spatial patterns. Effective for classification and feature extraction.	Computationally intensive. Requires large amounts of data.	Well-suited for detecting spatial patterns in network flows, such as packet structure, which might help in detecting anomalies based on patterns in packet data.
LSTM (Long Short-Term Memory)	Lower	0.98	Higher	Excellent for time-based pattern detection. Suitable for temporal sequence learning and anomaly detection.	Computationally slower. Difficult to train. Requires large amounts of data.	Good for detecting time-based patterns, such as sequences of abnormal network activity over time, which might occur in malicious behaviors (e.g., DDoS attacks, botnets).
Autoencoder	Lower	0.94	Higher	Suitable for anomaly detection. Unsupervised learning. Effective for data compression and reconstruction.	Poor performance in classification. Can suffer from reconstruction bias. Requires structure tuning.	Useful for detecting anomalies by reconstructing normal traffic patterns and flagging deviations (e.g., malicious activity).

On the other hand, LSTM albeit slightly inferior to CNN, outperformed all the other models in capturing temporal patterns especially in terms of high PR-AUC scores and the highest accuracy in IoT23 dataset making LSTM a model of choice for any task involving temporal sequence modeling. Despite being highly accurate in detecting anomalies the autoencoder fell short of the other performance parameters and hence were less effective for high accuracy tasks with balanced precision recall measurements. The summary of comparison results for IoT-23 dataset is presented in Table II.

CNNs and LSTMs can scale to large IoT environments, detecting threats in real-time to prevent early attacks. This research also supports developing self-learning models by combining CNNs, LSTMs, and Autoencoders for enhanced IoT security.

V. CONCLUSION

From the comparative analysis of the CNN, LSTM and autoencoder models for IoT intrusion detection systems using the IoT-23 dataset, it was observed that CNN model provided the better PR-AUC and accuracy values than the other models. The CNN's improved ability to balance between the precision and the recall associated with a high accuracy indicates that it is the most appropriate for general pattern recognition in the network traffic. As can be seen from the results, LSTM with a slight loss to CNN showed a high performance in terms of learning temporal dependencies and therefore, could be recommended for use in tasks which involve the utilization of temporal data. The autoencoder proved useful in detecting anomalies but had relatively lower accuracy which means that the model is not very suitable for environments that call for greater precision and accuracy as well as optimal precision-recall. Thus, these results are useful for choosing an adequate model depending on specified conditions and evaluation metrics, which will help improve the utilization of IoT security applications.

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